



Machine Learning and the Physical World

Lecture 6 : Statistical Emulation

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29th October, 2025

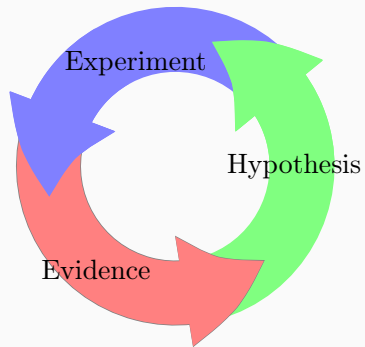
<http://carlhenrik.com>

- Why is uncertainty not just desirable but necessary for learning

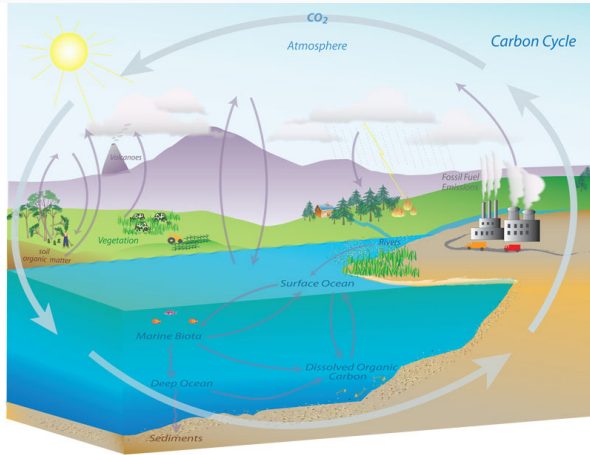
- Why is uncertainty not just desirable but necessary for learning
- Why any prediction can only be interpreted in the light of the beliefs that lead to them

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- Why any prediction can only be interpreted in the light of the beliefs that lead to them
- Mathematical formalism for computing with uncertainties
- Gaussian processes as a means of parametrising beliefs over functions



$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{\int p(y \mid \theta)p(\theta)d\theta}$$



Simulator

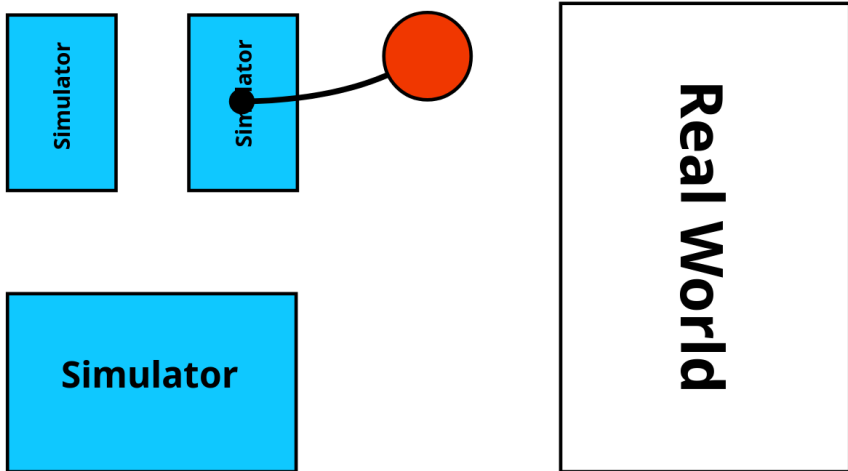
Simulator

Simulator

Real World

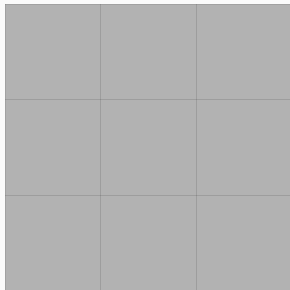
Types of Simulation

- Differential Equation Models
 - epidemiological models, climate/weather models
- Discrete Event Simulations
 - turn-based
 - event-driven
- Computer Code

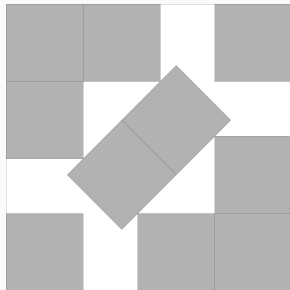


 **Statistical Emulators**

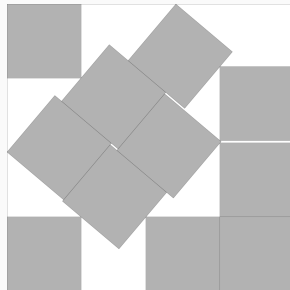
Packing Problem



$$s = 3$$

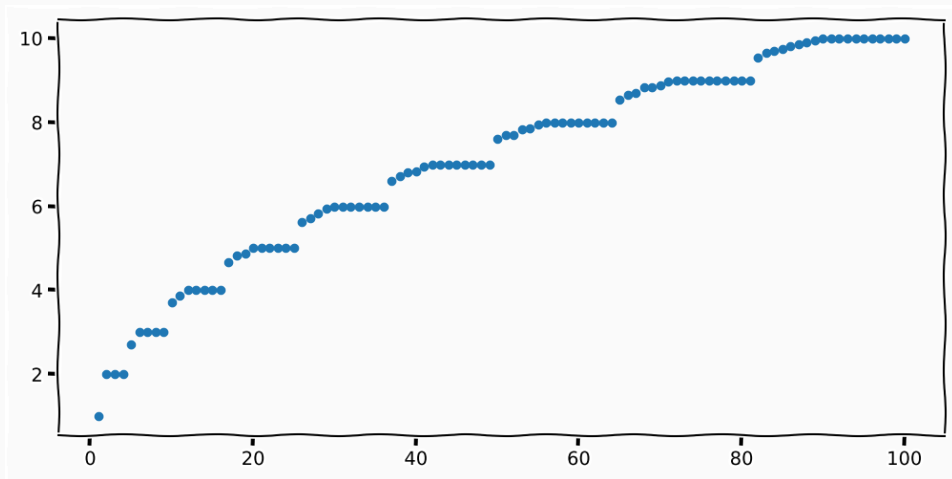


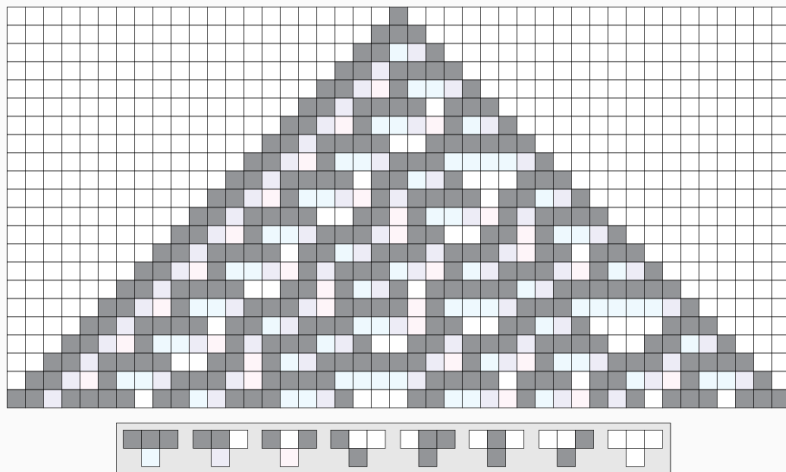
$$s = 3 + \frac{1}{2}\sqrt{2}$$

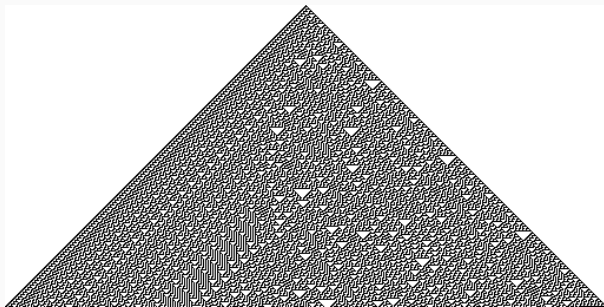


$$\begin{aligned} s^8 - 20s^7 + 178s^6 - \\ 842s^5 + 1923s^4 - \\ 496s^3 - 6754s^2 + \\ 12420s - 6865 = 0 \end{aligned}$$

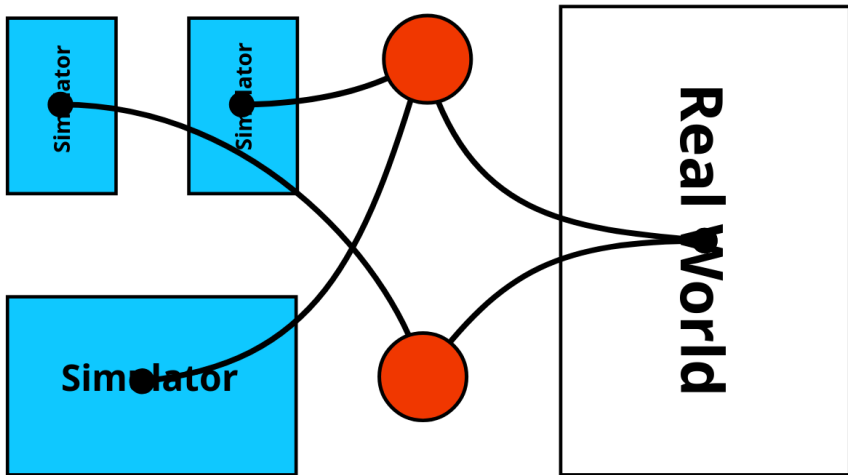
Data Regularity





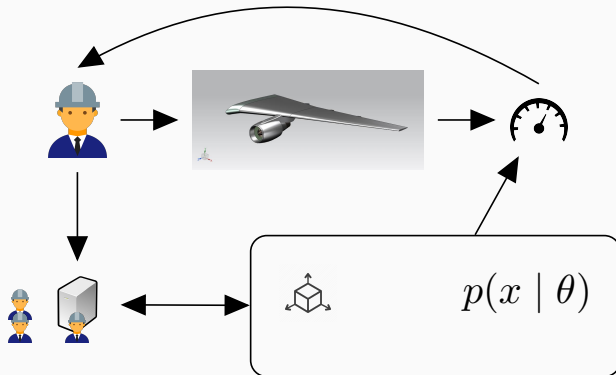


Current Pattern	1, 1, 1	1, 1, 0	1, 0, 1	1, 0, 0	0, 1, 1	0, 1, 0	0, 0, 1	0, 0, 0
New State	0	0	0	1	1	1	1	0

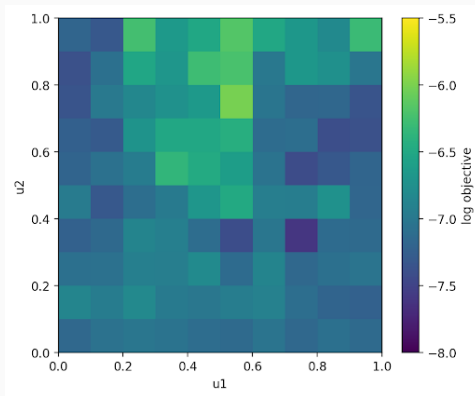
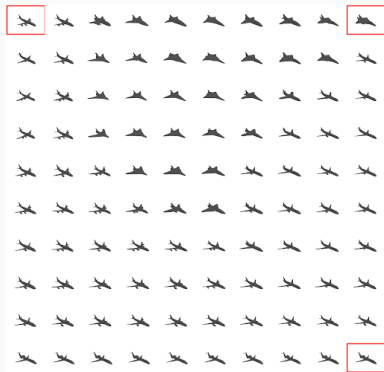


● Statistical Emulators

Boeing Example

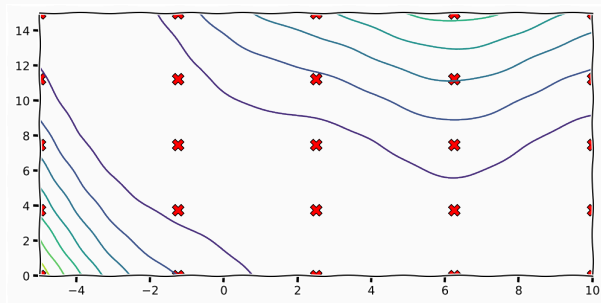


Boeing Example



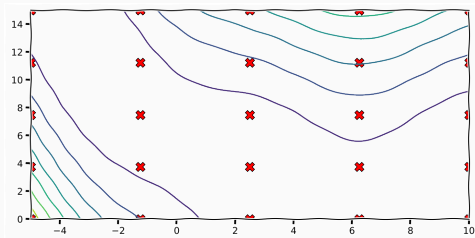
Statistical Emulation

Branin Function



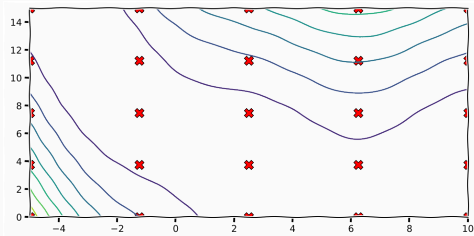
$$f(\mathbf{x}) = a(x_2 - bx_1^2 + cx_1 - r)^2 + s(1 - t \cos(x_1)) + s$$

Branin Function



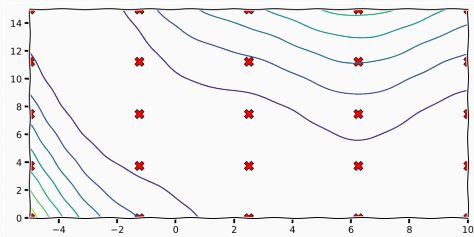
$$p(U) \sim \mathbb{U}[(-5, 10) \times (0, 15)]$$
$$\mathbb{E}[f(U)] = ?$$

Branin Function



$$\mathbb{E}[f(U)] \approx \frac{1}{n} \sum_i^N f(\mathbf{x}_i) = 73.5$$

Branin Function

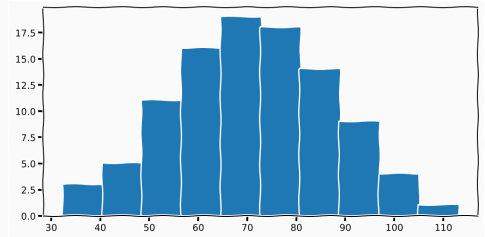
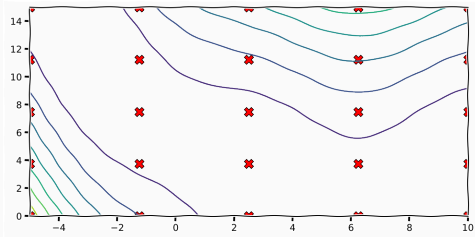


Fit a GP surrogate

$$k(\mathbf{x}_i, \mathbf{x}_j) = k_{\text{Exp}}(\mathbf{x}_i, \mathbf{x}_j) + k_{\text{White}}(\mathbf{x}_i, \mathbf{x}_j) + k_{\text{Bias}}$$

$$\mathbb{E}[f(U)] \approx \mathbb{E}[p(\mathbf{f} \mid \mathbf{Y}, \mathbf{X}, \mathbf{X}_{\text{samples}})] = 70.25$$

Branin Function





Questions that we want to answer with Emulation

- Where to query data next?

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- Where to query data next?
- What to sample next?

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- Where to query data next?
- What to sample next?
- How certain is my current estimate?

- Where to query data next?
- What to sample next?
- How certain is my current estimate?
- What is the most efficient way to reduce my uncertainty?

EmuKit

EmuKit/emukit

A Python-based toolbox of various methods in decision making, uncertainty quantification and statistical emulation: multi-fidelity, experimental design, Bayesian optimisation, Bayesian...



 32

Contributors

 98

Used by

 635

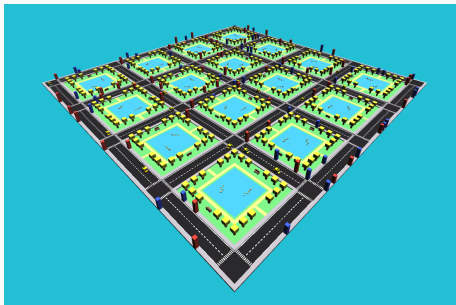
Stars

 131

Forks



<https://emukit.github.io/>



<https://emukit.github.io/emukit-playground/>

<https://proceedings.scipy.org/articles/gerudo-f2bc6f59-009>

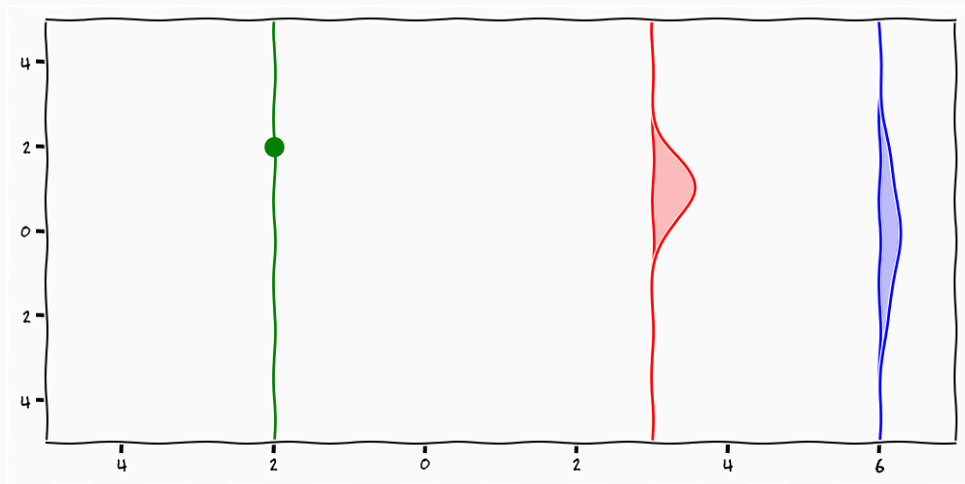
Gaussian Process Emulators

$$f \sim \mathcal{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

$$\mu(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$$

$$\text{cov}(f(\mathbf{x}), f(\mathbf{x}')) = k(\mathbf{x}, \mathbf{x}')$$

Gaussian Process: Sample Paths



$$\text{cov}(f(\mathbf{x}), f(\mathbf{x}')) = \mathbb{E}[(f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})])(f(\mathbf{x}') - \mathbb{E}[f(\mathbf{x}')])]$$

- Positive

$$k(\mathbf{x}, \mathbf{x}) = \text{cov}(f(\mathbf{x}), f(\mathbf{x})) = \text{var}(f(\mathbf{x})) \geq 0$$

- Symmetric

$$k(\mathbf{x}, \mathbf{x}') = \text{cov}(f(\mathbf{x}), f(\mathbf{x}')) = \text{cov}(f(\mathbf{x}'), f(\mathbf{x}))$$

What is a Kernel: Positive semi-definite

$$T = \sum_{i=1}^n a_i f(\mathbf{x}_i)$$

$$\begin{aligned} \text{var}(T) &= \text{cov} \left(\sum_{i=1}^n a_i f(\mathbf{x}_i), \sum_{j=1}^n a_j f(\mathbf{x}_j) \right) = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{cov}(f(\mathbf{x}_i), f(\mathbf{x}_j)) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j k(\mathbf{x}_i, \mathbf{x}_j) \geq 0 \end{aligned}$$

Definition (PSD)

A positive semi-definite function satisfies the above inequality for all $n \in \mathbb{N}$, $\mathbf{x}_i \in \mathcal{D}$ and $a_i \in \mathbb{R}$

Theorem (Loeve)

$k(\cdot, \cdot)$ is a covariance of a GP $\Leftrightarrow k(\cdot, \cdot)$ is a positive semi-definite function

Squared Exponential $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2\ell^2} \right)$

Matern 5/2 $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{5}|\mathbf{x} - \mathbf{x}'|}{\ell} + \frac{5|\mathbf{x} - \mathbf{x}'|^2}{3\ell^2} \right) \exp \left(-\frac{\sqrt{5}|\mathbf{x} - \mathbf{x}'|}{\ell} \right)$

Matern 3/2 $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{3}|\mathbf{x} - \mathbf{x}'|}{\ell} \right) \exp \left(-\frac{\sqrt{3}|\mathbf{x} - \mathbf{x}'|}{\ell} \right)$

Exponential $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(-\frac{|\mathbf{x} - \mathbf{x}'|}{\ell} \right)$

Rational Quadratic $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{(\mathbf{x} - \mathbf{x}')^2}{2\alpha\ell^2} \right)^{-\alpha}$

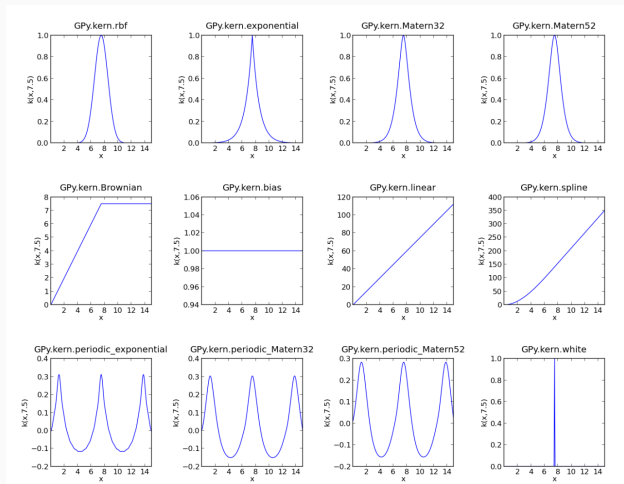
Periodic $k(\mathbf{x}, \mathbf{x}') = \exp \left(-\frac{2 \sin^2(\pi \frac{|\mathbf{x} - \mathbf{x}'|}{p})}{\ell^2} \right)$

Brownian $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \min(\mathbf{x}, \mathbf{x}')$

White $k(\mathbf{x}', \mathbf{x}) = \sigma^2 \delta_{\mathbf{x}, \mathbf{x}'}$

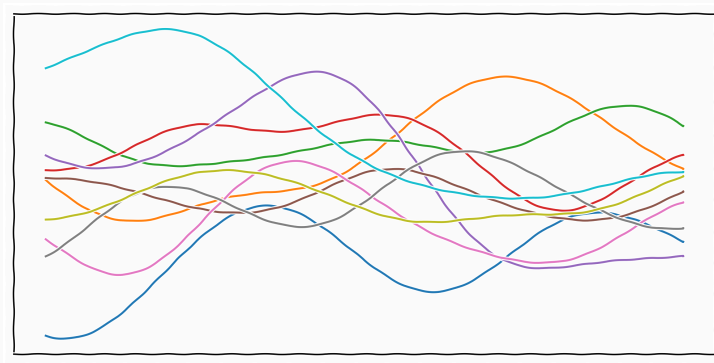
Constant $k(\mathbf{x}, \mathbf{x}') = \sigma^2$

Linear $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \mathbf{x}^T \mathbf{x}'$



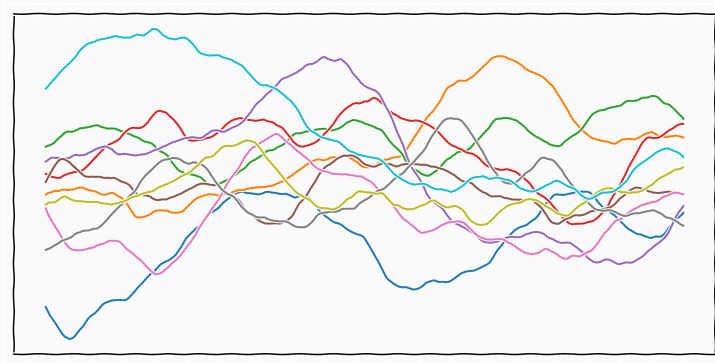
¹Taken from Nicola Durrande

Sample Paths: Exponentiated Quadratic



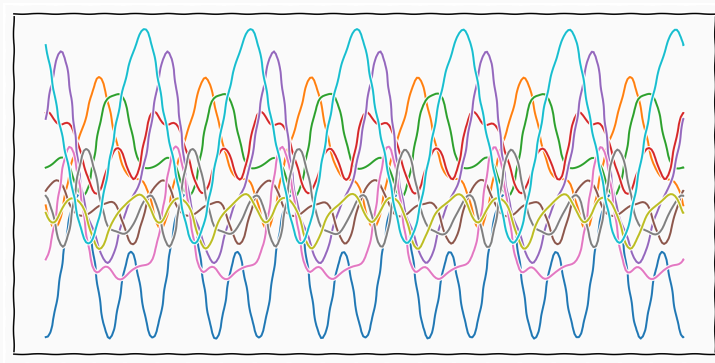
$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2\ell^2} \right)$$

Sample Paths: Matern 3/2



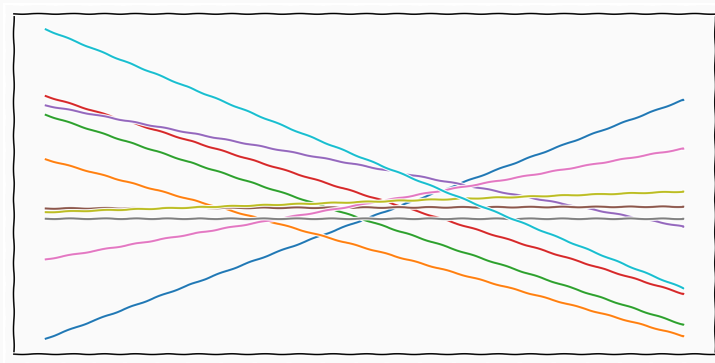
$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{3}|\mathbf{x} - \mathbf{x}'|}{\ell} \right) \exp \left(-\frac{\sqrt{3}|\mathbf{x} - \mathbf{x}'|}{\ell} \right)$$

Sample Paths: Periodic



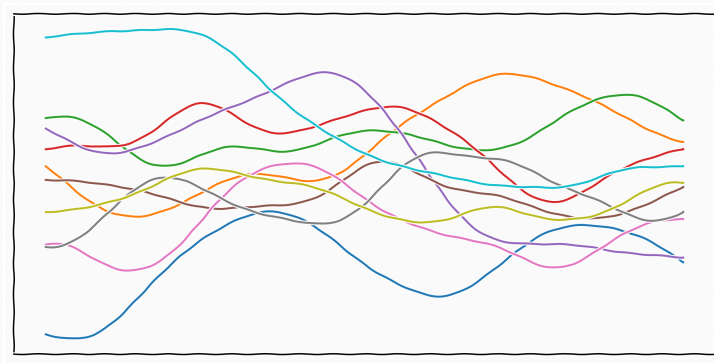
$$k(\mathbf{x}, \mathbf{x}') = \exp \left(-\frac{2 \sin^2 \left(\pi \frac{|\mathbf{x} - \mathbf{x}'|}{p} \right)}{\ell^2} \right)$$

Sample Paths: Linear



$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2\ell^2} \right)$$

Sample Paths: Rational Quadratic



$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp \left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2\ell^2} \right)$$

Theorem (Bochner)

A continous stationary function $k(x, y) = \tilde{k}(|x - y|)$ is positive definite if and only if $\tilde{k}(\cdot)$ is the Fourier transform of a finite positive measure

$$\tilde{k}(t) = \int_{\mathbb{R}} e^{-i\omega t} d\mu(\omega)$$

- Positive Measure

$$\mu(\omega) = \begin{cases} 1 & -1 < \omega < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Fourier Transform

$$\tilde{k}(t) = \frac{\sin(t)}{t}$$

- Valid kernel

$$k(x, y) = \frac{\sin(x - y)}{x - y}$$

- The Gaussian is the Fourier transform of itself
- Matern kernels are the Fourier transform of

$$\mu(\omega) = \frac{1}{(1 + \omega^2)^p}$$

- You can create very strange kernels by thinking of the spectral measure Wilson et al., [2013](#)

$$k(x, x') = ck_1(x, x')$$

$$k(x, x') = f(x)k_1(x, x')f(x')$$

$$k(x, x') = q(k_1(x, x'))$$

$$k(x, x') = \exp(k_1(x, x'))$$

$$k(x, x') = k_1(x, x') + k_2(x, x')$$

$$k(x, x') = k_1(x, x')k_2(x, x')$$

$$k(x, x') = k_3(\phi(x), \phi(x'))$$

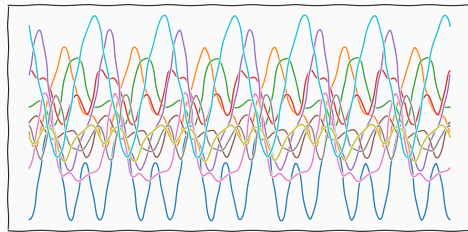
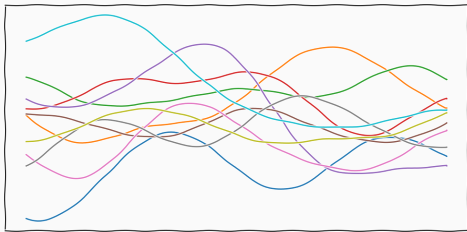
$$k(x, x') = x^T \mathbf{A} x'$$

$$k(x, x') = k_a(x_a, x'_a) + k_b(x_b, x'_b)$$

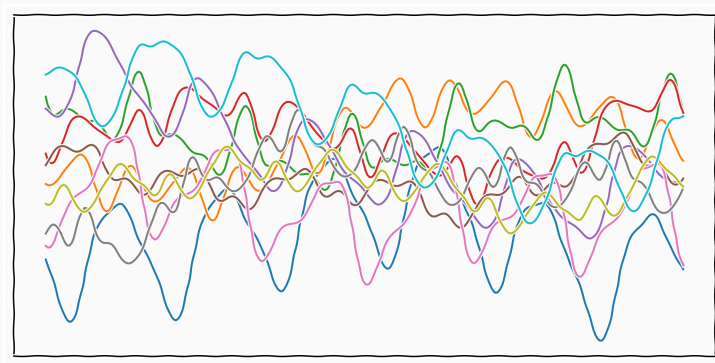
$$k(x, x') = k_a(x_a, x'_a)k_b(x_b, x'_b)$$

Positive definite functions are closed under certain operations

Combining Kernels

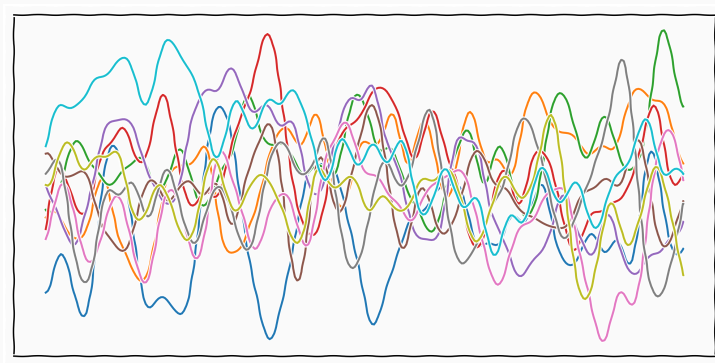


Combining Kernels: Addition



$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}')$$

Combining Kernels: Product



$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}')_2(\mathbf{x}, \mathbf{x}')$$

To choose the co-variance we need domain knowledge

- Is the process stationary?

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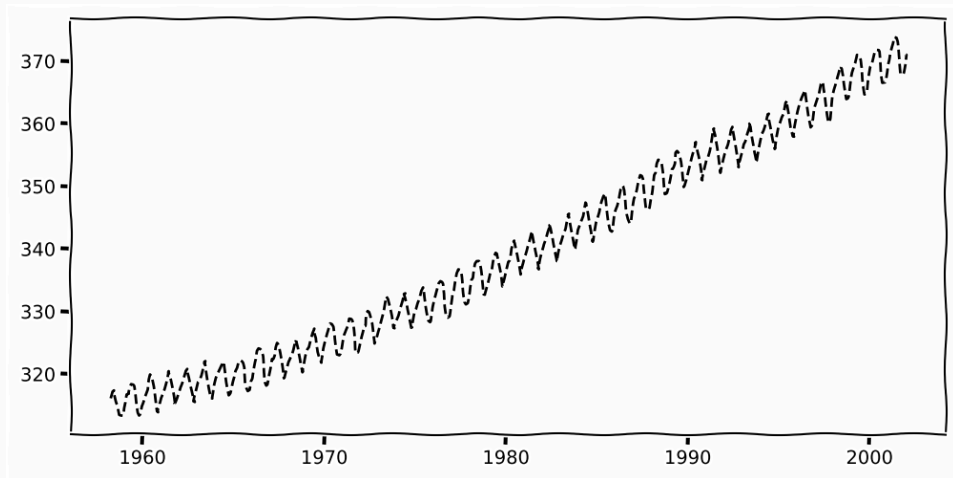
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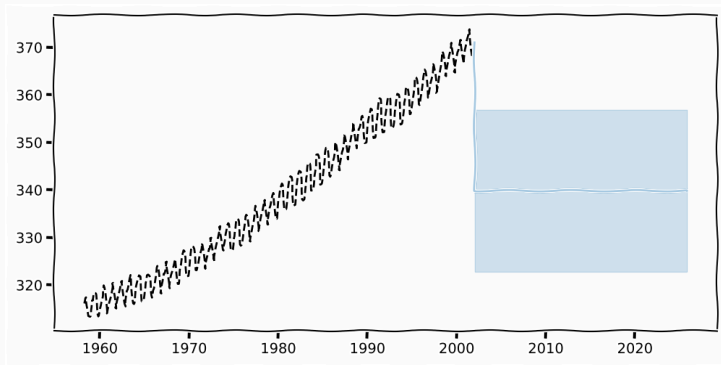
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- Do we expect trends in the data?
- Do we expect particular patterns
 - periodicity?
 - cycles?
 - additivity?
- Should we model in a transformed space?
 $x \rightarrow \log(x), x \rightarrow \exp(x), \dots$

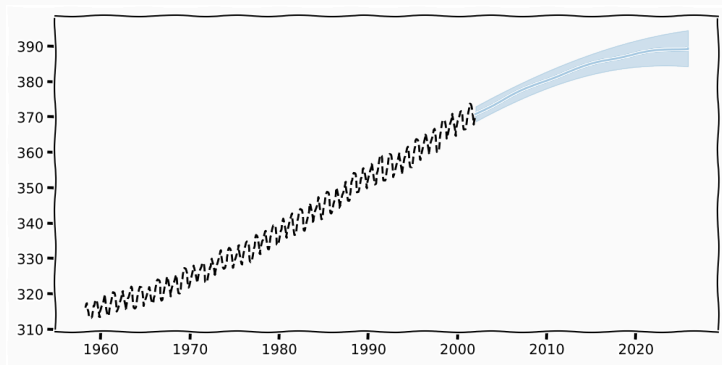
Mauna Loa (Rasmussen et al., 2006)



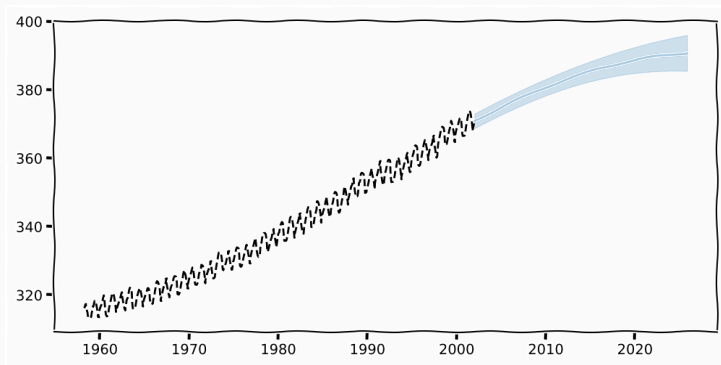
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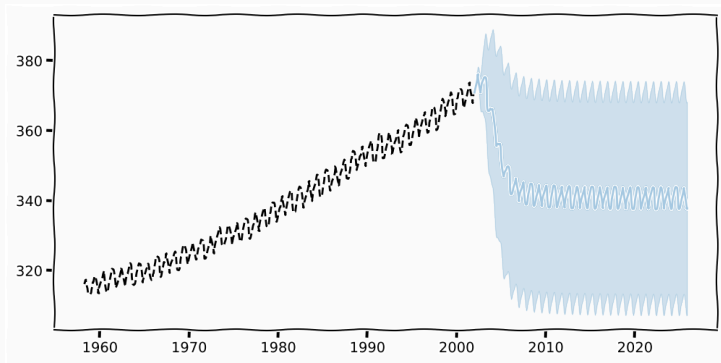
$$k(t, t') = k_{\text{SQE}}(t, t')$$



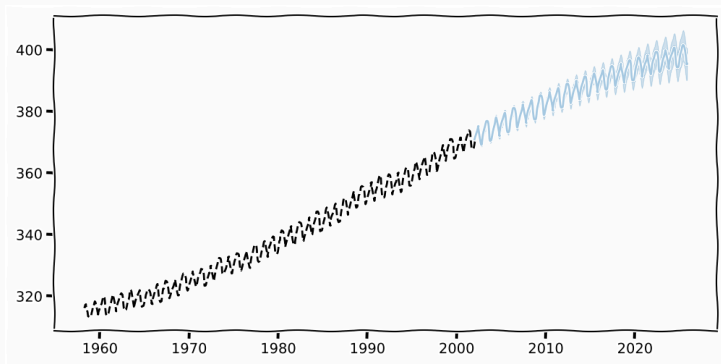
$$k(t, t') = k_{\text{SQE}}(t, t') + k_{\text{White}}(t, t')$$



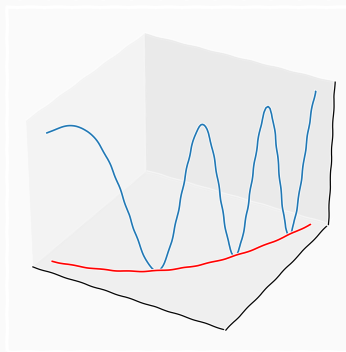
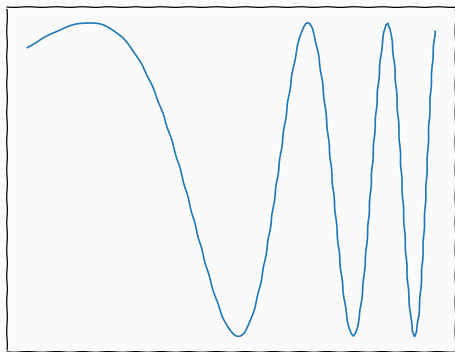
$$k(t, t') = k_{\text{SQE}}(t, t') + k_{\text{White}}(t, t') + k_{\text{Periodic}}(t, t')$$



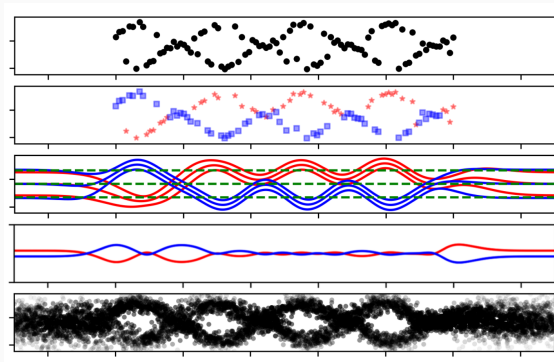
$$k(t, t') = k_{\text{SQE}}(t, t') + k_{\text{White}}(t, t') + k_{\text{Periodic}}(t, t') + k_{\text{RBF}}(t, t') + k_{\text{RQ}}(t, t')$$



$$k(t, t') = k_{\text{SQE}}(t, t') + k_{\text{SQE}}(t, t') \cdot k_{\text{Periodic}}(t, t') + k_{\text{RQ}}(t, t') + k_{\text{White}}(t, t')$$

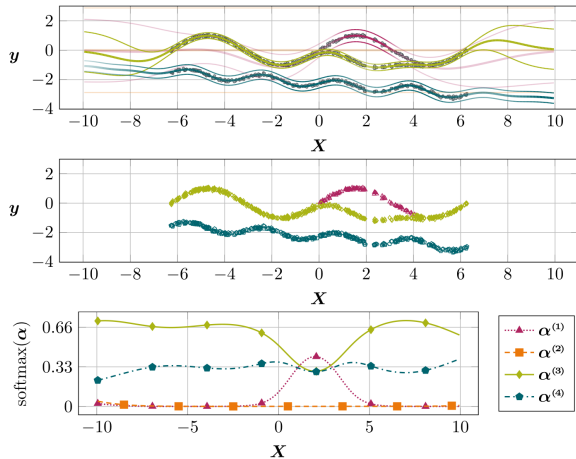


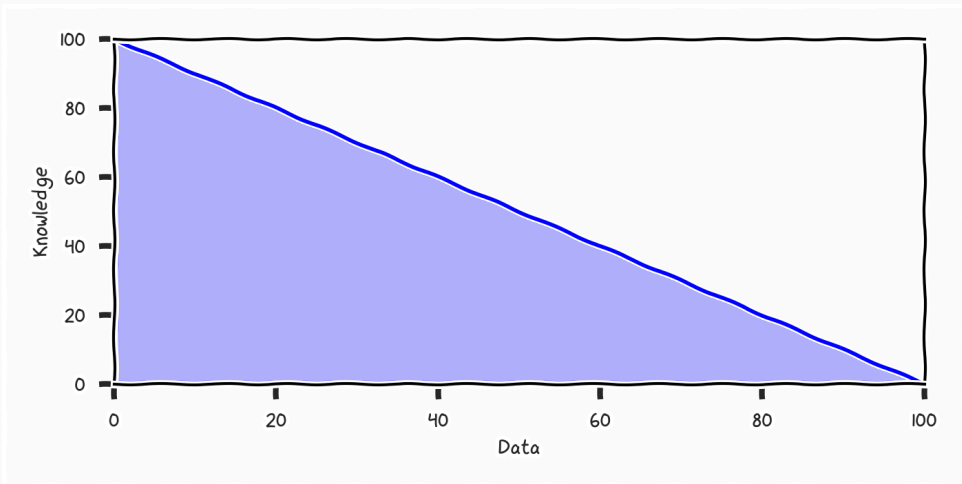
$$k((\mathbf{x}, \mathbf{x}_l), (\mathbf{x}', \mathbf{x}'_l))$$



$$k((\mathbf{x}, \mathbf{x}_l), (\mathbf{x}', \mathbf{x}'_l))$$

Switching Processes (Kaiser et al., 2018)







Summary

- What is an emulator?
- How do simulations and emulation interact?
- Gaussian process emulators
 - very rich framework for inputting domain knowledge

eof

References

References

-  Bishop, Christopher M. (2006). *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Secaucus, NJ, USA: Springer-Verlag New York, Inc.
-  Bodin, Erik, Neill D. F. Campbell, and Carl Henrik Ek (2017). “**Latent Gaussian Process Regression**”. In: *CoRR*.
-  Kaiser, Markus, Clemens Otte, Thomas Runkler, and Carl Henrik Ek (2018). “**Data Association With Gaussian Processes**”. In: *CoRR*.

-  Rasmussen, Carl Edward and Christopher K I Williams (2006). *Gaussian Processes for Machine Learning (Adaptive Computation and Machine Learning)*. The MIT Press.
-  Wilson, Andrew and Ryan Adams (2013). “Gaussian Process Kernels for Pattern Discovery and Extrapolation”. In: *Proceedings of the 30th International Conference on Machine Learning*. Ed. by Sanjoy Dasgupta and David McAllester. Vol. 28. Proceedings of Machine Learning Research 3. Atlanta, Georgia, USA: PMLR, pp. 1067–1075.